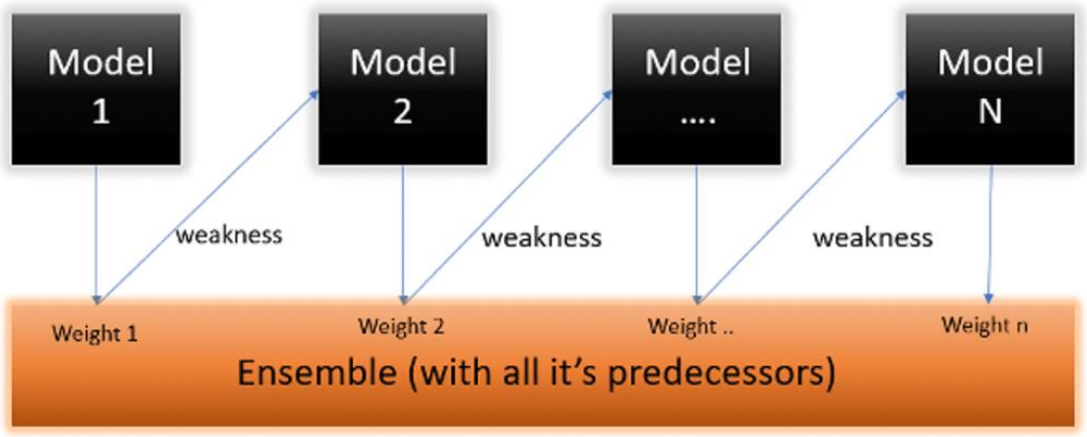
Boosting Algorithm

Ada Boost:

AdaBoost (Adaptive Boosting) is an ensemble learning algorithm that combines multiple weak learners to create a strong predictive model.



Advantages and disadvantages

Advantages:

Simple to understand and implement

Combines many weak learners effectively

Disadvantages:

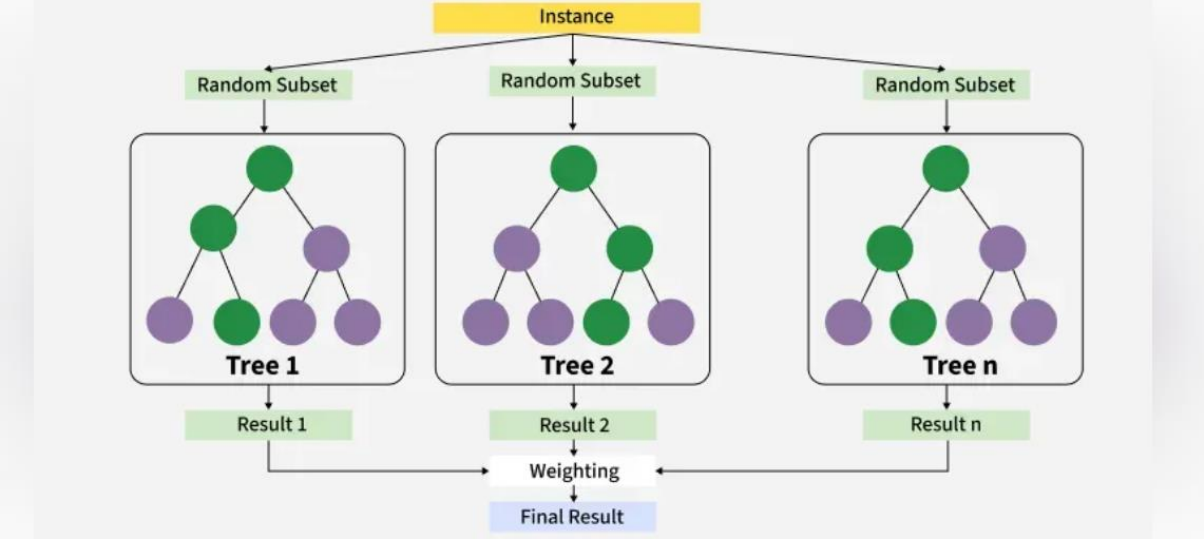
Sensitive to noisy data and outliers

Can overfit in some situations

XG Boost:

XGBoost is an ensemble learning algorithm that combines multiple decision trees to make accurate predictions. Each new tree improves the model by correcting errors made by the existing trees.

Boosting Algorithm



Advantages and disadvantages

Advantages:

High accuracy on many tabular datasets

Handles nonlinear relationships

Disadvantages:

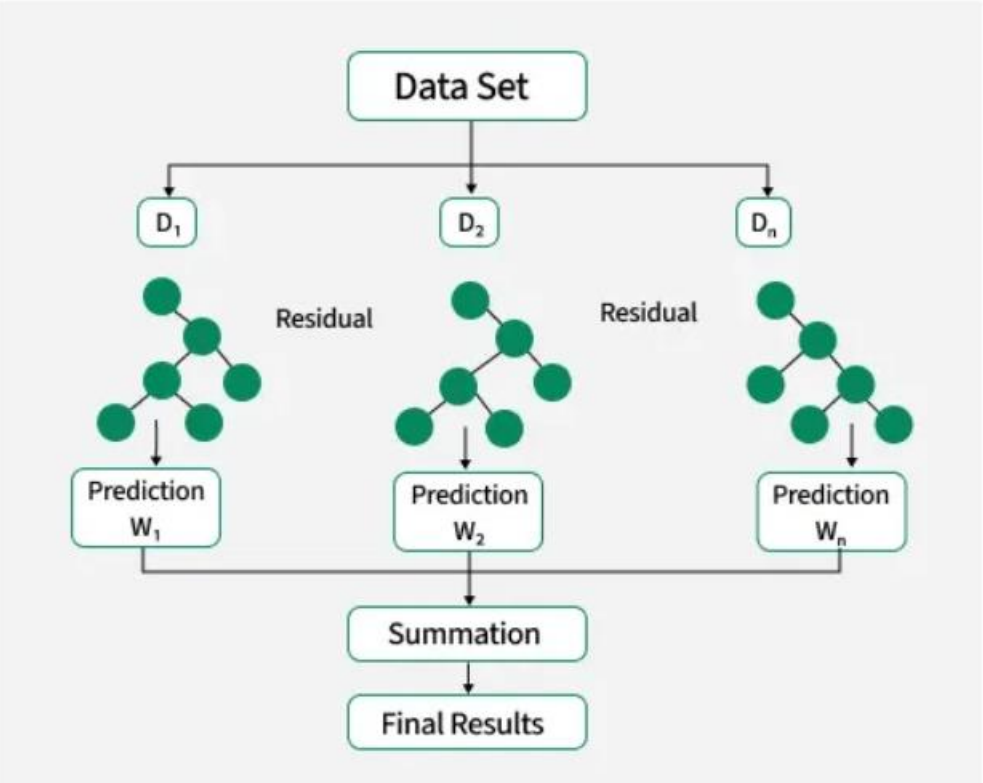
Requires hyperparameter tuning

Can overfit if poorly tuned

LG Boosting:

LightGBM is a tree-based ensemble learning algorithm. It builds multiple decision trees sequentially, with each new tree helping reduce the prediction errors of the existing model.

Boosting Algorithm



Advantages:

Speed and efficiency

Handles large datasets with fast training and low memory usage